13 EXAM PRACTICE

1 An equilibrium is represented by the following equation:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g); \quad \Delta H^{\ominus} = -92 \text{ kJ mol}^{-1}$$

Which change would affect both the value of the equilibrium constant, K_p, and the proportion of ammonia present at equilibrium?

A adding a catalyst of finely divided iron

B reducing the temperature at constant pressure

C increasing the amount of nitrogen

D increasing the pressure at constant temperature [1]

(Total for Question 1 = 1 mark)

2 In a reversible reaction, what is the effect of adding a catalyst on the rate constant k_1 for the forward reaction, the rate constant k_{-1} for the backward reaction and on the equilibrium constant K?

	k_1	k_{-1}	K
A	increases	decreases	no effect
B	increases	decreases	increases
C	increases	increases	no effect
D	increases	increases	increases

[1]

(Total for Question 2 = 1 mark)

3 Hydrogen and iodine vapour exist in equilibrium with hydrogen iodide at a constant temperature in a gas syringe:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

What will increase when the pressure is increased at constant temperature?

A the activation energy of the reaction

B the enthalpy change of the reaction

C K_p

D the partial pressure of hydrogen iodide [1]

(Total for Question 3 = 1 mark)

4 A nitrogen–hydrogen mixture, initially in the mole ratio of 1:3, reaches equilibrium with ammonia when 50% of the nitrogen has reacted.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The total pressure of the equilibrium mixture is P.

What is the partial pressure of ammonia in the equilibrium mixture?

A $P/6$ **B** $P/4$ **C** $P/3$ **D** $P/2$ [1]

(Total for Question 4 = 1 mark)

5 Nitrogen dioxide decomposes on heating according to the following equation:

$$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$$

When 4.0 mol of nitrogen dioxide is put into a 1 dm^3 container and heated, the equilibrium mixture obtained contains 0.8 mol of oxygen.

What is the numerical value of the equilibrium constant, K_c, at the temperature of the experiment?

A $\dfrac{0.8 \times 0.8}{2.4}$ **B** $\dfrac{(0.8)^2 \times 0.8}{2.4}$

C $\dfrac{1.6 \times 0.8}{(2.4)^2}$ **D** $\dfrac{(1.6)^2 \times 0.8}{(2.4)^2}$ [1]

(Total for Question 5 = 1 mark)

6 This question is about the equilibrium:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H^{\ominus} = -92 \text{ kJ mol}^{-1}$$

Which of the following statements is *not* correct?

A The units of K_p are atm^{-2}.

B K_p increases when the temperature decreases.

C K_p increases when the pressure increases.

D K_p increases when the total entropy change, ΔS_{total}, increases. [1]

(Total for Question 6 = 1 mark)

7 Nitrogen and hydrogen react together to form ammonia in a reversible reaction that can reach a position of equilibrium. The equation for the reaction is:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g); \qquad \Delta H_r = -92 \text{ kJ mol}^{-1}$$

At a fixed temperature and a total pressure of 5.00 atm, a sealed vessel of volume 20.0 dm^3 contained 1.00 mol of $N_2(g)$, 2.00 mol of $H_2(g)$ and 1.00 mol of $NH_3(g)$.

(i) Calculate the value, stating units, of the equilibrium constant, K_c, for this reaction. [5]

(ii) Calculate the value, stating units, of the equilibrium constant, K_p, for this reaction. [5]

(Total for Question 7 = 10 marks)

8 When phosphorus(V) chloride, PCl_5, is heated to a constant temperature in a sealed vessel, it forms a gaseous equilibrium mixture with phosphorus(III) chloride, PCl_3, and chlorine. The equation for the reaction is:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

(a) The equilibrium constant may be expressed in terms of either concentration, K_c, or partial pressure, K_p.

(i) Write an expression for K_p. [1]

(ii) If the units of partial pressure are atm, state the units of K_p for the above equilibrium. [1]

(b) Explain, in terms of the expression for K_p, the effect on the position of equilibrium of increasing the pressure at constant temperature. [4]

(c) The relationship between K_c and K_p is given by the expression:

$$K_p = K_c (0.0821\,T)^{\Delta n}$$

where T = the absolute temperature in kelvin (K) and

Δn = the change in number of moles from reactants to products.

Calculate a value for K_c if the value for K_p at 500 K is 0.810 atm. [3]

(Total for Question 8 = 9 marks)

9 A chemist has discovered a method for making a commercially important chemical, R. The reaction involves the following reversible reaction:

$$2P(g) + Q(g) \rightleftharpoons R(g)$$

The process is normally carried out in industry at a pressure of 500 atm and a temperature of 573 K.

(a) (i) Write an expression for K_p for this reaction. [1]

(ii) Give the units of K_p in this expression. [1]

(b) During their initial research, the chemist carried out several experiments, all at 500 atm pressure, using mixtures of P and Q, starting with 2.0 mol of P and 1.0 mol of Q. In each case they allowed the reaction mixture to reach equilibrium before determining the percentage of Q converted.

The table gives the results of two of his experiments.

	Experiment 1	Experiment 2
Temperature/K	423	573
Percentage of Q converted	50	70

(i) Calculate, using the results from experiment 1, the value of K_p. [4]

(ii) Deduce the conditions of temperature and pressure that would give the highest yield of R. [2]

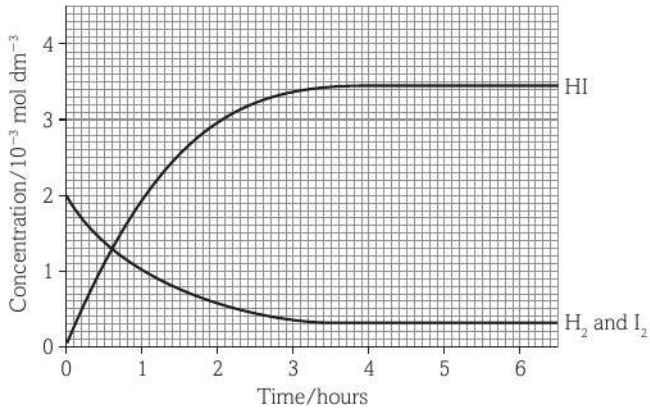
(iii) Explain why the conditions you have suggested in part (ii) may be different from those used industrially. [2]

(Total for Question 9 = 10 marks)

10 Hydrogen and iodine react together, in the presence of a suitable catalyst, in a reversible reaction. The equation for the reaction is:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g); \qquad \Delta H^{\ominus} = -9\,\text{kJ mol}^{-1}$$

(a) The graph shows the changes in concentration when 2.0×10^{-3} mol of H_2 were mixed with 2.0×10^{-3} mol of I_2 in a sealed container of volume 1 dm^3 and left to reach equilibrium.



(i) Write the expression for the equilibrium constant, K_c, for this reaction. [1]

(ii) Use the graph to determine the time at which equilibrium was established. [1]

(iii) Use the graph to determine the equilibrium concentrations of HI(g), H_2(g) and I_2(g). [2]

(iv) Calculate a value for the equilibrium constant, K_c. State the units. [2]

(b) Explain how the value of K_c changes when:

(i) the total pressure at equilibrium is increased at constant temperature [2]

(ii) the temperature is increased at constant pressure. [2]

(Total for Question 10 = 10 marks)

11 The composition of an equilibrium mixture produced at 2 atm pressure is shown here:

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$

0.80	0.80	1.20	3.60	Amount in equilibrium mixture/mol

(a) Give the expression for the equilibrium constant, K_p, for the reaction and calculate its value. Include units in your answer. [6]

(b) The total entropy change, in J mol^{-1} K^{-1}, is related to the equilibrium constant by the equation

$$\Delta S^{\ominus}_{\text{total}} = R \ln K$$

Calculate the total entropy change for this reaction.

(R = 8.31 J mol^{-1} K^{-1}) [1]

(Total for Question 11 = 7 marks)